Mohammad Amin Roshani

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Scholar | LinkedIn | GitHub

Education

Wayne State University 2023 – Present
Ph.D. in Computer Science

Wayne State University 2023 – 2025
M.S. in Computer Science

The University of Memphis 2022 – 2023
Ph.D. studies in Computer Science (transferred to Wayne State) *GPA: 3.94*

The University of Tehran 2017 – 2022
B.Sc. in Computer Science *GPA: 3.1*

National Organization for Development of Exceptional Talents 2013 – 2017
Mathematics High School Diploma *GPA: 4.0*

Interests

Health Informatics, Machine Learning, Natural Language Processing, Data Analysis

Experience

Graduate Student Instructor Jan 2026 – May 2026
Wayne State University *Instructor: Dongxiao Zhu*

- CSC 5991: Trustworthy AI for Large Language Models and Vision-Language Models
- Led weekly labs on LLM/VLM foundations, supervised fine-tuning, prompting, CLIP/LLaVA, red-teaming, unlearning, grounded VLMs (LLaVA+SAM), and agentic AI safety
- Mentored student teams designing, evaluating, and documenting trustworthy AI applications

Graduate Teaching Assistant Aug 2025 – May 2026
Wayne State University

- Problem Solving and Programming – Hadi Nasser

Graduate Teaching Assistant Aug 2024 – May 2025
Wayne State University

- Introduction to Web Technology – Kamel Rushaidat
- Senior Capstone Project – Seyed Ziae Mousavi Mojab

Graduate Research Assistant May 2023 – Aug 2024
Wayne State University *Supervisor: Dongxiao Zhu*

- Working on NIH project about using LLMs to build a conversational app to predict COVID-19 severity risk
- Conducting research on safety of LLMs and multimodal foundational models

Graduate Teaching Assistant Aug 2022 – Apr 2023
The University of Memphis

- CS1: Introduction to Programming (Python) – William Baggett
- Foundations of Computing – Thomas Watson

Private Tutor Feb 2020 – Aug 2022
Self-Employed

- Teaching Introduction to Python to students in Iran and Australia

Teaching Assistant Jan 2020 – Jan 2022
The University of Tehran

- Advanced Information Retrieval (NLP) – Hamidreza Keshavarz
- Fundamentals of Computer Science and Programming – Deniz Tahmouresi
- Machine Language and Assembly – Abbas Nowzari-Dalini
- Principles of Computer Systems – Bagher BabaAli

Publications

Learning to Poison Large Language Models for Downstream Manipulation Oct 2026
IEEE BigData 2026 (Short Paper)

Authors' Reply: Toward Retrieval-Grounded Evaluation for Conversational Large Language Model–Based Risk Assessment Mar 2026
JMIR AI

Automatic Calibration for Membership Inference Attack on Large Language Models Oct 2025
ECAI 2025

Generative Large Language Model–Powered Conversational AI App for Personalized Risk Assessment: Case Study in COVID-19 Mar 2025
JMIR AI

Languages

English (Fluent; TOEFL 106, GRE V 152), Persian (Native), Arabic (Elementary)

Projects

AI-powered web application for elderly health tracking (collaboration with nursing school) Dec 2025

Adding LLMs to the risk assessment app to make a conversational agent Mar 2024

Designing COVID-19 risk severity assessment for NIH project Oct 2023

Hate speech detection using Transfer Learning Oct 2022

Additional: Sarcasm detection, sentiment analysis, segmentation, Bi-LSTM hate speech, regression/statistics 2021–2022

Technical Skills

Programming Languages: Python (primary), C++, JavaScript, Go, R, SQL, Bash, HTML, CSS, LaTeX
Machine Learning & AI: Deep Learning, LLMs, Transformers, RAG, LoRA, prompt engineering, multi-agent AI, XAI, clinical NLP
ML Frameworks & Libraries: PyTorch, Hugging Face, Scikit-learn, NumPy, Pandas, TensorFlow, OpenCV, Matplotlib, NLTK
NLP: BERT, GPT-style, LLaMA, tokenization, instruction fine-tuning, information extraction from clinical text
Web & Application Development: React, Node.js, JavaScript, REST APIs, web and mobile-connected apps
Data Science & Modeling: Statistical modeling, regression, classification, experimental design, AUC/F1/precision-recall
Tools & Infrastructure: Git, Linux/Ubuntu, VS Code, GPU pipelines, Hugging Face ecosystem, shell scripting
Research & Writing: LaTeX, reproducible experiments, dataset construction, peer review

Honors

Admitted to the second level of the National Student Olympiad (Mathematics and Literature)
Top 5 of the class for the last four semesters
First place in University Dart Competitions

References

Available upon request.